

(12) **United States Patent**
Thompson et al.

(10) **Patent No.:** **US 12,400,269 B2**
(45) **Date of Patent:** ***Aug. 26, 2025**

(54) **SYSTEM AND METHOD FOR ADVERSE
VEHICLE EVENT DETERMINATION**

(71) Applicant: **Nauto, Inc.**, Palo Alto, CA (US)

(72) Inventors: **Alex Thompson**, Palo Alto, CA (US);
Shobana Sankaran, Palo Alto, CA
(US); **Charles Hornbaker**, Palo Alto,
CA (US); **Stefan Heck**, Palo Alto, CA
(US)

(73) Assignee: **Nauto, Inc.**, Sunnyvale, CA (US)

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 141 days.

This patent is subject to a terminal dis-
claimer.

(21) Appl. No.: **17/451,367**

(22) Filed: **Oct. 19, 2021**

(65) **Prior Publication Data**

US 2022/0036473 A1 Feb. 3, 2022

Related U.S. Application Data

(63) Continuation of application No. 16/552,320, filed on
Aug. 27, 2019, now Pat. No. 11,164,259, which is a
(Continued)

(51) **Int. Cl.**
G06Q 40/08 (2012.01)
G06N 7/01 (2023.01)

(Continued)

(52) **U.S. Cl.**
CPC **G06Q 40/08** (2013.01); **G06N 7/01**
(2023.01); **G06V 20/56** (2022.01); **B60R**
2021/01327 (2013.01)

(58) **Field of Classification Search**

None

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

5,584,035 A 12/1996 Duggan et al.
5,638,116 A 6/1997 Shumoura et al.
(Continued)

FOREIGN PATENT DOCUMENTS

DE 102009005730 7/2010
EP 3057061 8/2017

(Continued)

OTHER PUBLICATIONS

L. Mingsong, "Study on Traffic Accident Investigation and Site
Survey Evidence Based on Motor Vehicle Insurance," 2016 Eighth
International Conference on Measuring Technology and Mechatron-
ics Automation (ICMTMA), Macau, China, 2016, pp. 660-662.
(Year: 2016).*

(Continued)

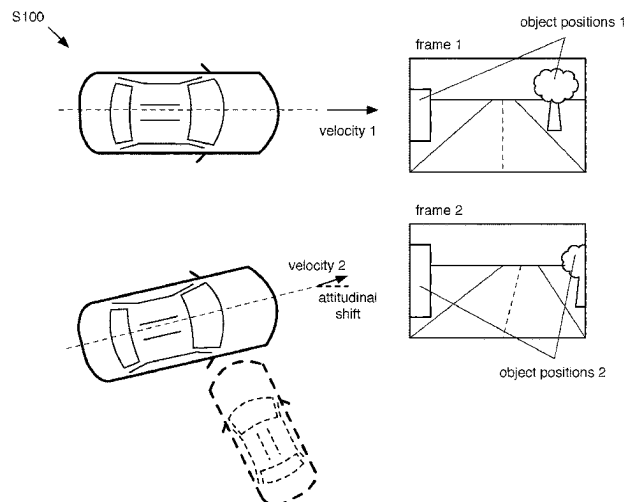
Primary Examiner — Mohammad Z Shaikh

(74) *Attorney, Agent, or Firm* — Vista IP Law Group,
LLP

(57) **ABSTRACT**

A method for determining an adverse vehicle event, includ-
ing: sampling sensor data an onboard vehicle system
coupled to an ego-vehicle; at the onboard vehicle system,
extracting a set of event parameters from the sensor data,
wherein the vehicle event data is associated with a vehicle
event occurring within the time interval; computing a loss
probability based on the set of event parameters in response
to the loss probability exceeding a threshold probability,
transforming the set of event parameters into insurance
claim data; and automatically transmitting the insurance
claim data to an endpoint, wherein the endpoint is deter-
mined based on the participant identifier.

20 Claims, 4 Drawing Sheets



Related U.S. Application Data

continuation of application No. 16/011,013, filed on Jun. 18, 2018, now Pat. No. 10,453,150.

- (60) Provisional application No. 62/521,058, filed on Jun. 16, 2017.

- (51) **Int. Cl.**
G06V 20/56 (2022.01)
B60R 21/0132 (2006.01)

- (56) **References Cited**

U.S. PATENT DOCUMENTS

5,642,106	A	6/1997	Hancock et al.	8,799,034	B1	8/2014	Brandmaier et al.
5,798,949	A	8/1998	Kaub	8,805,707	B2	8/2014	Schumann et al.
5,898,390	A	4/1999	Oshizawa et al.	8,854,199	B2	10/2014	Cook et al.
5,961,571	A	10/1999	Gorr et al.	8,862,380	B2	10/2014	Jung
6,018,728	A	1/2000	Spence et al.	8,934,709	B2	1/2015	Saptharishi et al.
6,240,367	B1	5/2001	Lin	8,952,819	B2	2/2015	Nemat-nasser
6,480,784	B2	11/2002	Mizuno	9,019,571	B2	4/2015	Yamada
6,496,117	B2	12/2002	Gutta et al.	9,053,554	B2	6/2015	Uchida et al.
6,502,033	B1	12/2002	Phuyal	9,079,571	B2	7/2015	Trost et al.
6,552,141	B1	4/2003	Chmelir et al.	9,081,650	B1	7/2015	Brinkmann et al.
6,662,141	B2	12/2003	Kaub	9,111,147	B2	8/2015	Thornton et al.
6,720,920	B2	4/2004	Breed et al.	9,121,713	B2	9/2015	Samarasekera et al.
6,927,694	B1	8/2005	Smith et al.	9,146,558	B2	9/2015	Field et al.
7,085,637	B2	8/2006	Breed et al.	9,158,962	B1	10/2015	Nemat-nasser et al.
7,148,913	B2	12/2006	Keaton et al.	9,180,887	B2	11/2015	Nemat-nasser et al.
7,177,737	B2	2/2007	Karlsson et al.	9,201,424	B1	12/2015	Ogale
7,195,394	B2	3/2007	Singh	9,201,932	B2	12/2015	Silver et al.
7,212,651	B2	5/2007	Viola et al.	9,218,003	B2	12/2015	Fong et al.
7,421,321	B2	9/2008	Breed et al.	9,235,750	B1	1/2016	Sutton et al.
7,423,540	B2	9/2008	Kisacanin	9,305,214	B1	4/2016	Young et al.
7,460,940	B2	12/2008	Larsson et al.	9,327,743	B2	5/2016	Green et al.
7,471,929	B2	12/2008	Fujioka et al.	9,330,571	B2	5/2016	Ferguson et al.
7,502,677	B2	3/2009	Weichenberger et al.	9,349,113	B2	5/2016	Bashkin
7,502,688	B2	3/2009	Hirokawa	9,358,976	B2	6/2016	Stierlin
7,551,093	B2	6/2009	Maass	9,412,102	B2	8/2016	Wolf et al.
7,558,672	B2	7/2009	Egami et al.	9,418,488	B1 *	8/2016	Lambert G07C 5/085
7,639,148	B2	12/2009	Victor	9,429,439	B2	8/2016	Stumper
7,646,922	B2	1/2010	Au et al.	9,439,036	B2	9/2016	Spears et al.
7,844,077	B2	11/2010	Kochi et al.	9,443,309	B2	9/2016	Menashe et al.
7,853,072	B2	12/2010	Han et al.	9,465,978	B2	10/2016	Hachisuka et al.
7,868,821	B2	1/2011	Hoshizaki	9,472,102	B2	10/2016	Mcclain et al.
7,912,288	B2	3/2011	Winn et al.	9,491,374	B1	11/2016	Avrahami et al.
7,954,587	B2	6/2011	Kisanuki et al.	9,514,626	B2	12/2016	Wu et al.
7,974,748	B2	7/2011	Goerick et al.	9,524,269	B1 *	12/2016	Brinkmann G07C 5/0808
8,022,831	B1	9/2011	Wood-Eyre	9,535,878	B1 *	1/2017	Brinkmann G08G 1/0112
8,073,287	B1	12/2011	Wechsler et al.	9,573,541	B2	2/2017	Graumann et al.
8,114,568	B2	2/2012	Van et al.	9,679,480	B2	6/2017	Hakeem
8,144,542	B2	3/2012	Na	9,688,150	B2	6/2017	Seong et al.
8,174,568	B2	5/2012	Samarasekera et al.	9,701,307	B1	7/2017	Newman et al.
8,195,394	B1	6/2012	Zhu et al.	9,718,468	B2	8/2017	Bartfield, Jr. et al.
8,254,670	B2	8/2012	Prokhorov	9,731,727	B2	8/2017	Heim et al.
8,266,132	B2	9/2012	Ofek et al.	9,734,414	B2	8/2017	Samarasekera et al.
8,301,344	B2	10/2012	Simon et al.	9,734,455	B2	8/2017	Levinson et al.
8,344,849	B2	1/2013	Larsson et al.	9,767,625	B1	9/2017	Snyder et al.
8,369,608	B2	2/2013	Gunaratne	9,812,016	B2	11/2017	Oremus
8,441,519	B2	5/2013	Oshima et al.	9,845,097	B2	12/2017	Prakah-Asante et al.
8,442,791	B2	5/2013	Stachlin	9,851,214	B1	12/2017	Chintakindi
8,447,519	B2	5/2013	Basnayake et al.	9,852,019	B2	12/2017	Ashani
8,487,775	B2	7/2013	Victor et al.	9,881,218	B2	1/2018	Ogata et al.
8,498,813	B2	7/2013	Oohashi et al.	9,892,558	B2	2/2018	Troy et al.
8,502,860	B2	8/2013	Demirdjian	9,928,432	B1	3/2018	Sathyaranayana et al.
8,510,196	B1	8/2013	Brandmaier et al.	9,977,973	B2	5/2018	Oluda et al.
8,594,920	B2	11/2013	Shida	10,078,333	B1	9/2018	Bajracharya
8,606,492	B1	12/2013	Botnen	10,176,524	B1 *	1/2019	Brandmaier G08G 1/0112
8,619,135	B2	12/2013	Shellshear et al.	10,387,962	B1 *	8/2019	Potter G08B 21/02
8,634,980	B1 *	1/2014	Urmson G06T 7/521	10,453,150	B2 *	10/2019	Thompson G06Q 50/40
			701/23	10,473,453	B2 *	11/2019	Kumagai G01B 11/00
8,654,151	B2	2/2014	Kim	11,162,800	B1 *	11/2021	Carbery G01C 21/28
8,666,644	B2	3/2014	Goto	11,164,259	B2 *	11/2021	Thompson G06Q 10/10
8,676,498	B2	3/2014	Ma et al.	11,436,844	B2 *	9/2022	Carruthers H04N 7/181
8,744,642	B2	6/2014	Nemat-nasser et al.	12,014,423	B1 *	6/2024	Mullen G08G 1/052
8,761,439	B1	6/2014	Kumar et al.	2001/0018636	A1	8/2001	Mizuno
				2002/0082806	A1	6/2002	Kaub
				2002/0140562	A1	10/2002	Gutta et al.
				2002/0180870	A1 *	12/2002	Chen H04N 5/144
							348/E5.065
				2002/0198632	A1	12/2002	Breed et al.
				2003/0095140	A1	5/2003	Keaton et al.
				2003/0169907	A1	9/2003	Edwards et al.
				2004/0051659	A1	3/2004	Garrison
				2004/0071319	A1 *	4/2004	Kikuchi G08G 1/052
							382/107
				2004/0167667	A1	8/2004	Goncalves et al.
				2004/0168148	A1	8/2004	Goncalves et al.
				2004/0258307	A1	12/2004	Viola et al.
				2005/0002558	A1	1/2005	Franke et al.
				2005/0060069	A1	3/2005	Breed et al.
				2005/0073136	A1	4/2005	Larsson et al.
				2005/0182518	A1	8/2005	Karlsson

(56)	References Cited			2013/0046510	A1 *	2/2013	Bowne	G06Q 10/0639 702/187
	U.S. PATENT DOCUMENTS			2013/0073114	A1	3/2013	Nemat-nasser et al.	
				2013/0093886	A1	4/2013	Rothschild	
				2013/0142390	A1	6/2013	Othmezzouri et al.	
				2013/0147661	A1	6/2013	Kangas et al.	
				2013/0151288	A1 *	6/2013	Bowne	G01S 19/13 705/4
2005/0234679	A1	10/2005	Karlsson	2013/0155229	A1	6/2013	Thornton et al.	
2005/0273218	A1	12/2005	Breed et al.	2013/0179198	A1 *	7/2013	Bowne	G07C 5/08 705/4
2005/0273263	A1	12/2005	Egami et al.	2013/0194127	A1	8/2013	Ishihara et al.	
2006/0106534	A1	5/2006	Kawamata et al.	2013/0211687	A1	8/2013	Trost et al.	
2006/0186702	A1	8/2006	Kisanuki et al.	2013/0226409	A1 *	8/2013	Akiyama	B60W 10/184 701/41
2006/0187305	A1	8/2006	Trivedi et al.	2013/0236107	A1 *	9/2013	Fukaya	G06T 7/246 382/201
2006/0187305	A1	8/2006	Trivedi et al.	2014/0037138	A1	2/2014	Sato et al.	
2006/0247847	A1	11/2006	Carter et al.	2014/0049601	A1	2/2014	Pfeil	
2006/0271287	A1	11/2006	Gold et al.	2014/0139655	A1	5/2014	Mimar	
2007/0043491	A1	2/2007	Goerick et al.	2014/0195477	A1	7/2014	Graumann et al.	
2007/0050108	A1	3/2007	Larschan et al.	2014/0210625	A1	7/2014	Nemat-Nasser	
2007/0063855	A1	3/2007	Maass	2014/0210978	A1	7/2014	Gunaratne et al.	
2007/0100669	A1	5/2007	Wargin et al.	2014/0213300	A1	7/2014	Spears et al.	
2007/0120948	A1	5/2007	Fujioka et al.	2014/0214255	A1	7/2014	Dolgov et al.	
2007/0154063	A1	7/2007	Breed	2014/0222280	A1	8/2014	Salomonsson et al.	
2007/0154100	A1 *	7/2007	Au	2014/0267703	A1	9/2014	Taylor et al.	
			G06V 20/56 382/226	2014/0297170	A1	10/2014	Sakima et al.	
2007/0159344	A1	7/2007	Kisacanin	2014/0300739	A1	10/2014	Mimar	
2007/0219720	A1 *	9/2007	Trepagnier	2014/0324281	A1	10/2014	Nemat-nasser et al.	
			G05D 1/024 701/300	2014/0379233	A1	12/2014	Chundrlík et al.	
2007/0244640	A1	10/2007	Hirokawa	2015/0006023	A1 *	1/2015	Fuchs	G06Q 30/0283 701/32.2
2007/0263901	A1	11/2007	Wu et al.	2015/0025917	A1	1/2015	Stempora	
2007/0280505	A1	12/2007	Breed et al.	2015/0049195	A1	2/2015	Ishigaki et al.	
2007/0291130	A1 *	12/2007	Broggi	2015/0078632	A1	3/2015	Hachisuka et al.	
			H04N 23/90 348/E13.015	2015/0084757	A1	3/2015	Annibale et al.	
2008/0025568	A1	1/2008	Han et al.	2015/0086078	A1	3/2015	Sibiryakov	
2008/0049975	A1 *	2/2008	Stiegler	2015/0110344	A1	4/2015	Okumura	
			G06T 7/20 382/104	2015/0127390	A1 *	5/2015	Bowne	G01C 21/3676 705/4
2008/0051957	A1	2/2008	Breed et al.	2015/0140991	A1	5/2015	Silver et al.	
2008/0075367	A1	3/2008	Winn et al.	2015/0149218	A1 *	5/2015	Bayley	B60R 21/0136 705/4
2008/0084283	A1	4/2008	Kalik	2015/0154845	A1	6/2015	Wu et al.	
2008/0167814	A1	7/2008	Samarasekera et al.	2015/0161892	A1	6/2015	Oremus	
2008/0243378	A1	10/2008	Zavoli	2015/0213556	A1 *	7/2015	Haller, Jr.	G06Q 40/08 705/4
2008/0252412	A1	10/2008	Larsson et al.	2015/0219462	A1	8/2015	Stumper	
2008/0260207	A1 *	10/2008	Nagaoka	2015/0221136	A1	8/2015	Shaburova et al.	
			G06T 7/32 382/103	2015/0228077	A1	8/2015	Menashe et al.	
2009/0080697	A1	3/2009	Kishikawa et al.	2015/0239482	A1	8/2015	Green et al.	
2009/0175498	A1	7/2009	Kochi et al.	2015/0254603	A1	9/2015	Bashkin	
2009/0244291	A1	10/2009	Saptharishi et al.	2015/0269438	A1	9/2015	Samarasekera et al.	
2009/0261979	A1	10/2009	Breed et al.	2015/0274161	A1	10/2015	Stierlin	
2010/0061591	A1	3/2010	Okada et al.	2015/0284001	A1	10/2015	Watanabe et al.	
2010/0100276	A1 *	4/2010	Fujinawa	2015/0294422	A1	10/2015	Carver et al.	
			H04N 23/68 348/148	2015/0307048	A1 *	10/2015	Santora	B60R 21/013 348/148
2010/0169013	A1	7/2010	Nakamura et al.	2015/0344030	A1	12/2015	Damerow et al.	
2010/0176987	A1	7/2010	Hoshizaki	2015/0363656	A1 *	12/2015	Brauer	G06V 20/56 348/77
2010/0209881	A1	8/2010	Lin et al.	2015/0375756	A1	12/2015	Do et al.	
2010/0214411	A1 *	8/2010	Weinmann	2015/0379715	A1	12/2015	Chandrasekar et al.	
			G06V 20/59 348/148	2016/0063341	A1	3/2016	Ogata et al.	
2010/0215254	A1	8/2010	Prokhorov	2016/0063761	A1	3/2016	Sisbot et al.	
2010/0217524	A1	8/2010	Oohashi et al.	2016/0078303	A1	3/2016	Samarasekera et al.	
2010/0225665	A1	9/2010	Ofek et al.	2016/0086021	A1	3/2016	Grohman et al.	
2010/0238009	A1	9/2010	Cook et al.	2016/0139977	A1	5/2016	Ashani	
2010/0312745	A1	12/2010	Tabak	2016/0147230	A1	5/2016	Munich et al.	
2010/0322507	A1	12/2010	Gunaratne	2016/0152211	A1 *	6/2016	Owens	B60R 25/102 348/148
2011/0128374	A1	6/2011	Shellshear et al.	2016/0163198	A1	6/2016	Dougherty	
2011/0137527	A1	6/2011	Simon et al.	2016/0169690	A1	6/2016	Bogovich et al.	
2011/0245993	A1	10/2011	Goto	2016/0176397	A1	6/2016	Prokhorov et al.	
2011/0262004	A1	10/2011	Murakami	2016/0203373	A1	7/2016	Menashe et al.	
2011/0301779	A1	12/2011	Shida	2016/0209511	A1	7/2016	Dolinar et al.	
2011/0316980	A1	12/2011	Dubbelman et al.	2016/0244022	A1	8/2016	Lippman et al.	
2012/0027258	A1	2/2012	Uchida et al.					
2012/0072243	A1 *	3/2012	Collins					
			G06Q 10/10 705/4					
2012/0078510	A1	3/2012	Ma et al.					
2012/0116676	A1	5/2012	Basnayake et al.					
2012/0121161	A1	5/2012	Eade et al.					
2012/0122486	A1	5/2012	Day et al.					
2012/0123806	A1	5/2012	Schumann, Jr. et al.					
2012/0154425	A1	6/2012	Kim					
2012/0185091	A1	7/2012	Field et al.					
2012/0197519	A1	8/2012	Richardson					
2012/0206596	A1	8/2012	Samarasekera et al.					
2012/0263346	A1	10/2012	Datta et al.					
2013/0006675	A1 *	1/2013	Bowne					
			G01S 19/13 705/4					

(56)

References Cited**U.S. PATENT DOCUMENTS**

2016/0253806 A1 9/2016 Iimura
 2016/0253886 A1 9/2016 Buchholz et al.
 2016/0267335 A1 9/2016 Hampiholi
 2016/0284078 A1 9/2016 Kim et al.
 2016/0297365 A1 10/2016 Nix
 2016/0297449 A1 10/2016 Heim et al.
 2016/0300242 A1 10/2016 Truong et al.
 2016/0305794 A1 10/2016 Horita et al.
 2016/0335475 A1 11/2016 Krenzer et al.
 2016/0339782 A1 11/2016 Seong et al.
 2017/0011529 A1 1/2017 Urashita
 2017/0039848 A1 2/2017 Hakeem
 2017/0039850 A1 2/2017 Vanden Berg et al.
 2017/0043781 A1 2/2017 Prakah-Asante et al.
 2017/0048239 A1 2/2017 Jeon et al.
 2017/0053167 A1 2/2017 Ren et al.
 2017/0053461 A1* 2/2017 Pal G08G 1/012
 2017/0053555 A1 2/2017 Angel et al.
 2017/0055868 A1 3/2017 Hatakeyama
 2017/0061222 A1 3/2017 Hoyer et al.
 2017/0064363 A1 3/2017 Wexler et al.
 2017/0080900 A1 3/2017 Huennekens et al.
 2017/0084048 A1* 3/2017 Tasaki G06T 7/73
 2017/0088142 A1 3/2017 Hunt et al.
 2017/0089710 A1 3/2017 Slusar
 2017/0098131 A1 4/2017 Shashua et al.
 2017/0101093 A1 4/2017 Barfield et al.
 2017/0106869 A1 4/2017 Lavoie et al.
 2017/0109828 A1 4/2017 Pierce et al.
 2017/0113664 A1 4/2017 Nix
 2017/0124476 A1 5/2017 Levinson et al.
 2017/0140231 A1 5/2017 Chen et al.
 2017/0144671 A1* 5/2017 Memani G06Q 40/12
 2017/0146801 A1 5/2017 Stempora
 2017/0178352 A1 6/2017 Harmsen et al.
 2017/0200061 A1 7/2017 Julian et al.
 2017/0221149 A1 8/2017 Hsu-Hoffman et al.
 2017/0221283 A1* 8/2017 Pal G06Q 40/08
 2017/0232963 A1* 8/2017 Pal G08B 25/00
 280/801.1
 2017/0243399 A1 8/2017 Troy et al.
 2017/0248952 A1 8/2017 Perkins et al.
 2017/0253236 A1 9/2017 Hayakawa
 2017/0287163 A1 10/2017 Kaufmann et al.
 2017/0292848 A1 10/2017 Nepomuceno et al.
 2017/0293819 A1 10/2017 Deng
 2017/0309072 A1 10/2017 Li et al.
 2017/0314954 A1 11/2017 Golding et al.
 2017/0345161 A1 11/2017 Takatani et al.
 2017/0357861 A1 12/2017 Okuda et al.
 2018/0012085 A1 1/2018 Blayvas et al.
 2018/0032891 A1* 2/2018 Ba B60W 50/14
 2018/0039862 A1 2/2018 Hyatt et al.
 2018/0043829 A1 2/2018 Cordell et al.
 2018/0045519 A1 2/2018 Ghadiok et al.
 2018/0052515 A1 2/2018 Wanner et al.

2018/0075309 A1 3/2018 Sathyanarayana et al.
 2018/0107882 A1 4/2018 Ogata et al.
 2018/0115711 A1 4/2018 Kato et al.
 2018/0172454 A1 6/2018 Ghadiok et al.
 2018/0176173 A1 6/2018 Keyzers et al.
 2018/0186366 A1 7/2018 Gordon et al.
 2018/0204111 A1 7/2018 Zadeh et al.
 2018/0229770 A1 8/2018 Kataoka et al.
 2018/0232583 A1 8/2018 Wang et al.
 2018/0239144 A1 8/2018 Woods et al.
 2018/0259353 A1 9/2018 Tsurumi et al.
 2019/0283763 A1* 9/2019 Wang G07C 5/08
 2021/0272208 A1* 9/2021 Leise G06Q 40/08

FOREIGN PATENT DOCUMENTS

GB 2506365 4/2014
 WO WO 2014154771 10/2014
 WO WO 2014/207558 A2* 12/2014 G06F 19/00
 WO WO 2015184578 12/2015
 WO WO 2016135561 9/2016
 WO WO 2016179303 11/2016
 WO WO 2018039560 3/2018

OTHER PUBLICATIONS

V. Krishnamoorthy, S. Agarwal and K. S. Venkatesh, "Vehicle trajectory prediction using a catadioptric omnidirectional camera," 2016 International Conference on Advances in Computing, Communications and Informatics (ICACCI), Jaipur, India, 2016, pp. 2761-2764. (Year: 2016).
 Non-Final Office Action for U.S. Appl. No. 16/552,320 dated Feb. 23, 2021.
 Notice of Allowance for U.S. Appl. No. 16/552,320 dated Jun. 29, 2021.
 Wei, Lijun, et al. "GPS and Stereovision-Based Visual Odometry: Application to Urban Scene Mapping and", International Journal of Vehicular Technology, vol. 24, No. 5, Article ID 439074, Jan. 11, 2011, 17 pages.
 Feng, Gu, et al. "Task 3—Evaluating the Relationship Between Near-Crashes and Crashes", Can Near-Crashes Serve as a Surrogate Safety Metric for Crashes? Virginia Tech Transportation Institute, U.S. Department of Transportation, Sep. 2010.
 "Which P&C Insurers Have Filed Patents Related to Autonomous Vehicles" Dec. 14, 2016, <https://www.cbinsights.com/research/autonomous-vehicle-insurancepatents/?ReillyBrennanFoT>, downloaded from the internet on Sep. 4, 2018.
 Notice of Allowance for U.S. Appl. No. 16/011,013 dated Jun. 12, 2019.
 Non-Final Office Action for U.S. Appl. No. 16/011,013 dated Jan. 23, 2019.
 International Search Report and the Written Opinion of the International Searching Authority, Application No. PCT/IB2018/000759, dated Nov. 30, 2018.

* cited by examiner

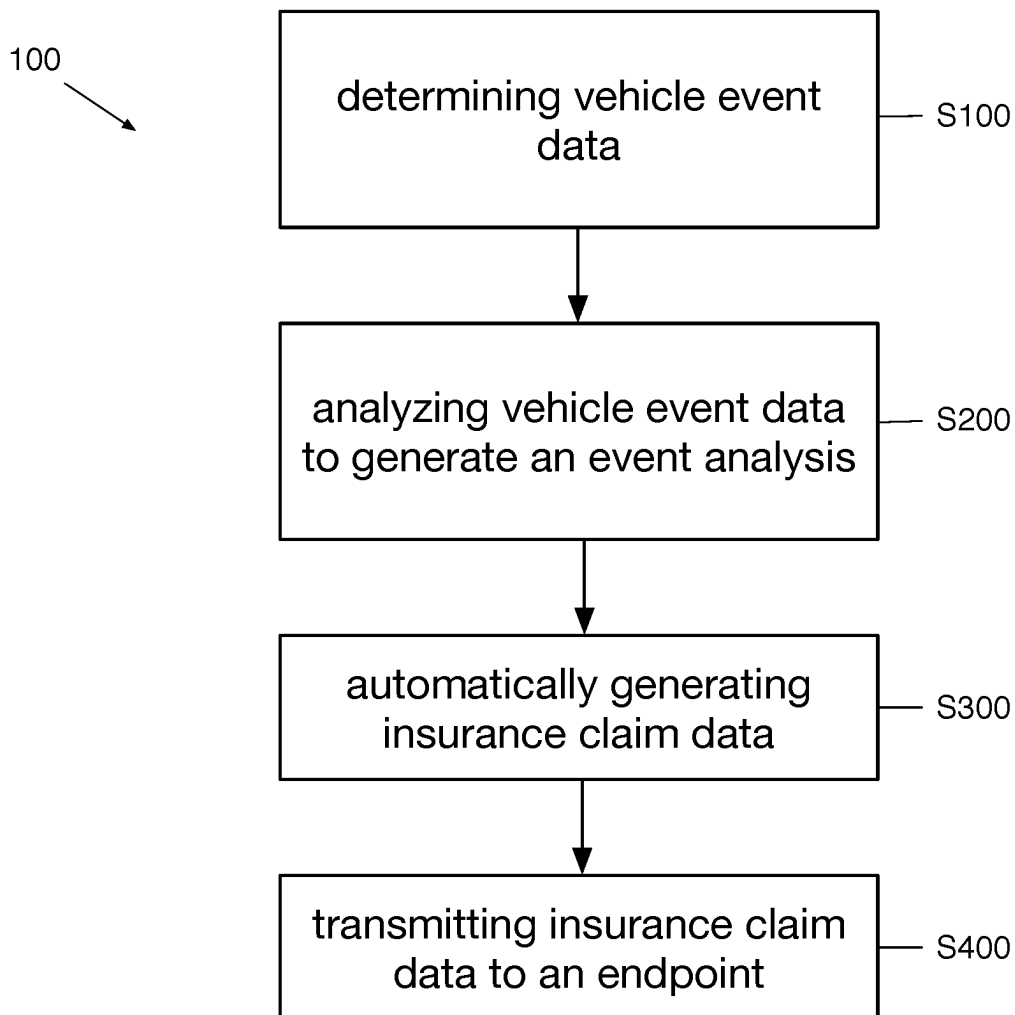


FIGURE 1

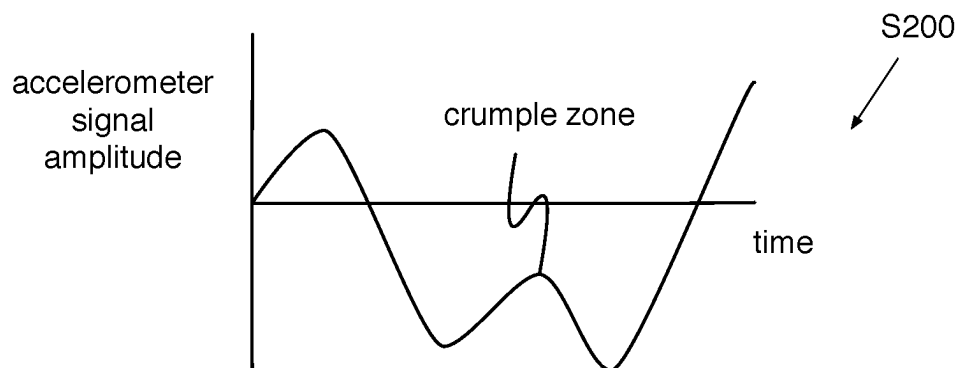


FIGURE 2

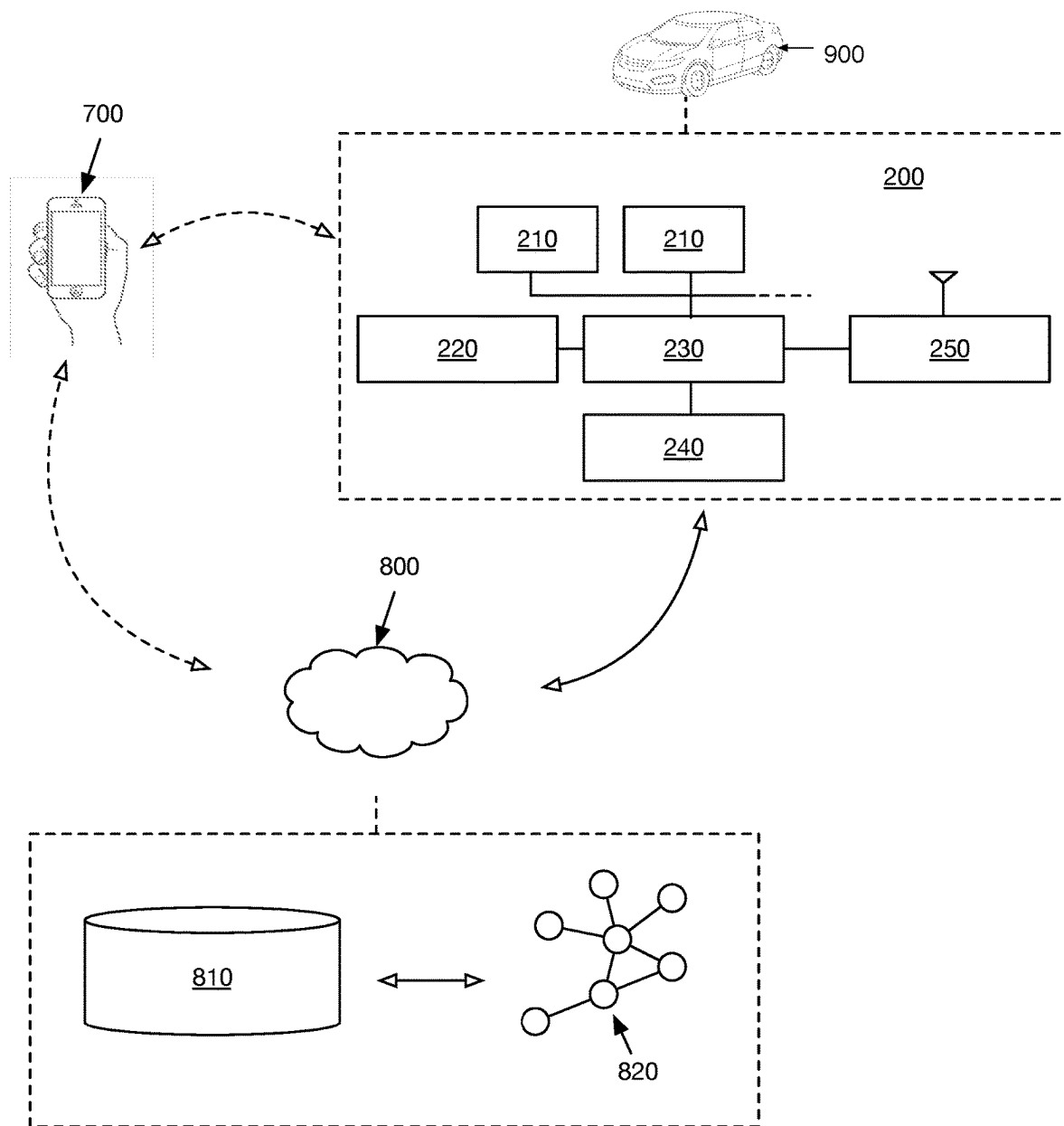


FIGURE 3

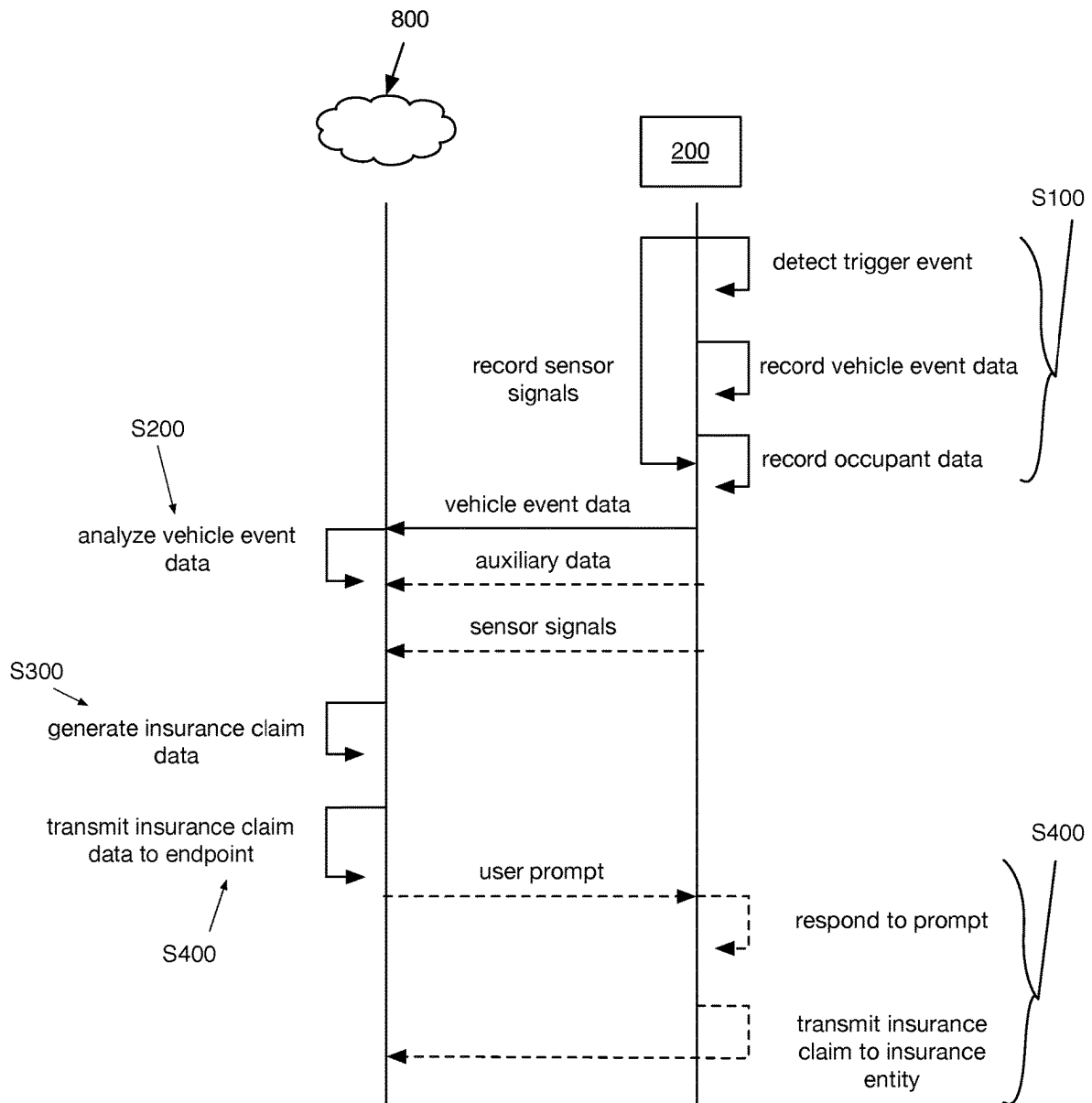


FIGURE 4

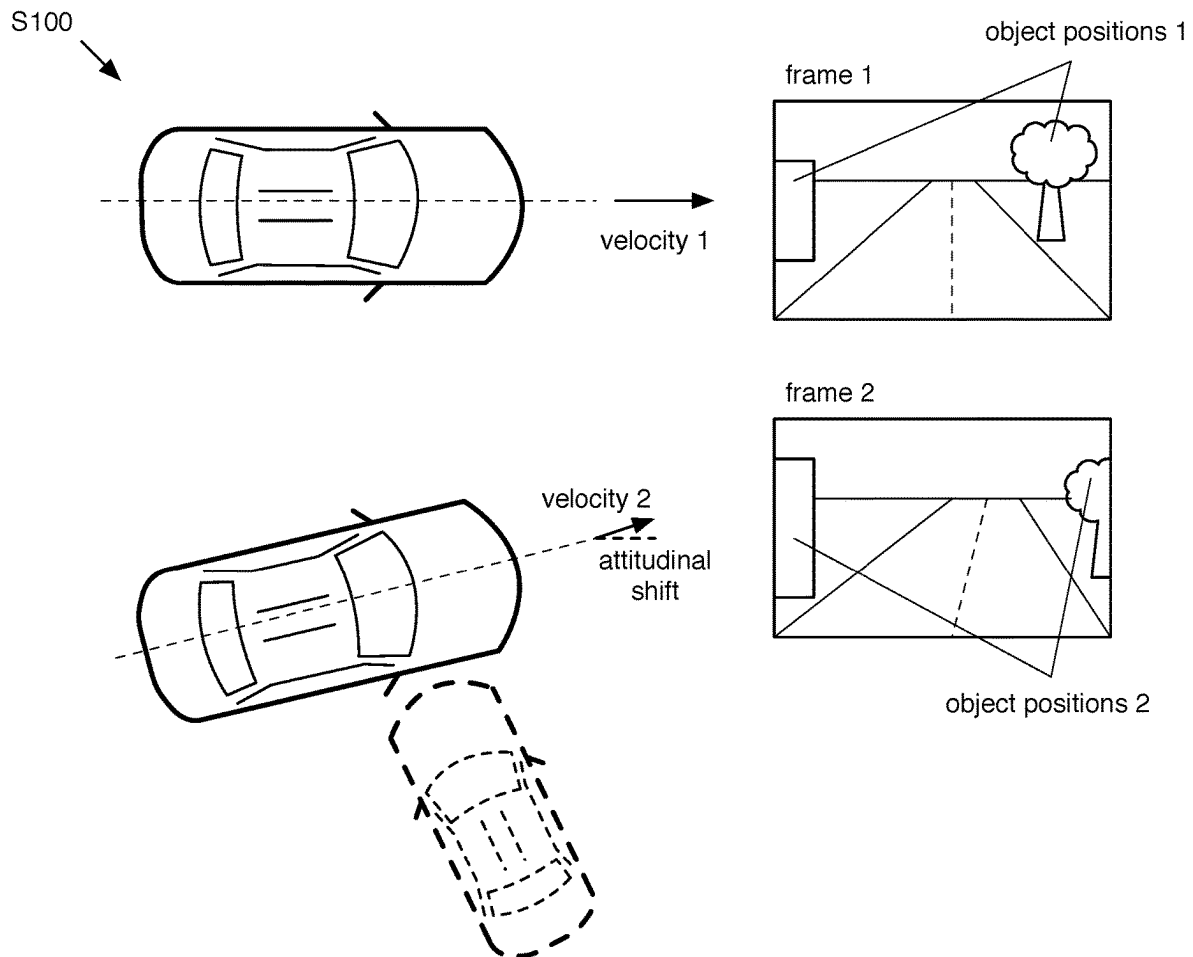


FIGURE 5

1

SYSTEM AND METHOD FOR ADVERSE VEHICLE EVENT DETERMINATION

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 16/552,320 filed on Aug. 27, 2019, now U.S. Pat. No. 11,164,259, which is a continuation of U.S. patent application Ser. No. 16/011,013 filed Jun. 18, 2018, now U.S. Pat. No. 10,453,150, which claims the benefit of U.S. Provisional Patent Application No. 62/521,058 filed Jun. 16, 2017. This application is related to U.S. patent application Ser. No. 15/892,899, filed Feb. 9, 2018. The entireties of all of the above-identified applications are expressly incorporated by reference herein

TECHNICAL FIELD

This invention relates generally to the automotive analysis field, and more specifically to a new and useful system and method for adverse vehicular event determination in the automotive analysis field.

BRIEF DESCRIPTION OF THE FIGURES

FIG. 1 is a flowchart representation of a variation of the method of adverse vehicle event determination.

FIG. 2 is an example signal curve indicative of an adverse vehicle event in an example implementation of the method.

FIG. 3 is a flowchart representation of example interrelated systems associated with an example implementation of the method.

FIG. 4 is a flowchart representation of a specific example implementation of the method.

FIG. 5 depicts a diagram of a specific implementation of a portion of the method.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

The following description of the preferred embodiments of the invention is not intended to limit the invention to these preferred embodiments, but rather to enable any person skilled in the art to make and use this invention.

1. Overview

As shown in FIG. 1, the method for adverse vehicle event determination includes: determining vehicle event data; analyzing vehicle event data to generate an event analysis; automatically generating insurance claim data based on the event analysis; and transmitting insurance claim data to an endpoint.

The method functions to determine that an adverse vehicle event has occurred. The method can also function to determine characteristics of the adverse vehicle event (e.g., severity, anticipated monetary cost, etc.). The method can also function to generate insurance claim data based on the characteristics of the adverse vehicle event. The method can also function to provide the insurance claim data (e.g., data that can be used to open or file an insurance claim, a first notice of loss/FNOL, etc.) to an entity (e.g., an agent at an insurance company, a vehicle-history monitoring service, etc.).

The method is preferably performed in response to occurrence of a vehicle event (e.g., a sharp change in the vehicle

2

attitude, impact of the vehicle with another object, receipt of a notification of vehicle damage, receipt of new sensor information, determination of vehicle event data, etc.). However, the method can additionally or alternatively be performed continuously in real- or near-real time, asynchronously, or at any other suitable time. In variations, the method is performed iteratively at a predetermined frequency (e.g., once per millisecond), performed a single time during a time period defined between a vehicle ignition time point and a vehicle turning-off time point, performed a single time for the vehicle, and/or be performed at any suitable frequency. In further variations, the method is performed after verification from a user (e.g., a crash is detected, followed by transmission of verification notifications to one or more user devices, followed by receipt of verification by the user, and finally followed by generation of insurance claim data). However, performance of the method can have otherwise suitable temporal characteristics.

As shown in FIG. 3, the method and/or variations thereof can be performed by and/or implemented at various related systems. The method is preferably performed related to a physical vehicle 900 traversing through a physical volume, but can additionally or alternatively be performed for a virtual model (e.g., of a vehicle) or otherwise suitably performed. The physical vehicle can be an automobile, a motorcycle, a bicycle, a skateboard, an aerial system (e.g., a drone, plane, etc.), or be any other suitable vehicle. The vehicle can be driven by a human driver, be automatically controlled, be telematically controlled (e.g., by a teleoperator), or be otherwise suitably controlled. The method is preferably performed for a single vehicle, but can additionally or alternatively be performed for a plurality of vehicles, a pair of vehicle involved in a vehicle event (e.g., in a crash), and/or any other suitable number and/or grouping of vehicles. The vehicle 900 is preferably equipped with an onboard vehicle system 200. The onboard vehicle system 200 preferably includes one or more sensors 210 (e.g., permanently integrated with the vehicle, installed in a removable dashboard-mounted module, etc.), a power system 220, an onboard processing system 230 (e.g., a vehicle computer), a memory system 240 (e.g., onboard FLASH memory storage), and a communication module 250. However, the vehicle can include any other suitable subsystems and/or components, either integrated into the vehicle (e.g., during manufacturing, as an aftermarket add on, etc.), coupled to a removable module (e.g., a mobile device, a dashboard-mounted system, etc.), and/or otherwise suitably associated with the vehicle. Variations of the method are preferably performed at least partially at a remote computing system 800 (e.g., a remote server), which can include one or more databases 810 (e.g., for storing data and/or from which to retrieve data) and analysis modules 820 (e.g., for performing any Blocks of the method and/or variations and examples thereof). Various blocks of the method can be performed in cooperation with a mobile device 700 associated with an occupant of the vehicle (e.g., an occupant device, a user device) or with another entity. However, the method and/or portions thereof can be otherwise suitably performed with any suitable system(s).

2. Benefits

Variants of the systems and/or methods can confer several benefits and/or advantages.

First, variants of the method can include automatically generating insurance claim data, including first notice of loss (FNOL) messages (e.g., reports), to simplify and streamline

3

handling and management of insurance claims. For example, the method can reduce the cost of collecting and organizing necessary information for an FNOL. The method can also enable timely provision of FNOLs to insurance entities (e.g., agencies, agents, underwriters, etc.) by reducing and/or eliminating the need for a human adjuster to assess the claim information. The method can also enable more accurate claim adjustment (e.g., by providing high quality data, such as video data, with the FNOL in order to determine characteristics of an adverse vehicle event leading to the FNOL such as accident fault, severity, additional parties involved, behavior of the driver, presence of injury of driver and/or other occupants, other risk events leading up to or otherwise proximal in time to the collision, etc.).

Second, variants of the method can enable automatic identification of parties involved in an adverse vehicle event (e.g., associating participant or party identities or identifiers with insurance claim data, simplifying and/or streamlining claim routing, enhancing claim routing efficiency, etc.). The method can include identifying a driver of the vehicle (e.g., using an interior camera that records interior video and/or images, using user profile data associated with the driver, using driver behavior signature pattern matching, etc.), identifying one or more passengers of the vehicle (e.g., using anonymized or non-anonymized interior imagery, data links with mobile devices of vehicle passengers, etc.), identifying other vehicular participants in an adverse vehicle event (e.g., a second vehicle involved in a crash with the primary vehicle, via exterior video data, image recognition, license-plate identification, etc.), identifying occupants of other vehicular participants of an adverse vehicle event (e.g., via facial recognition algorithms implemented on exterior imagery collected by the primary vehicle), and/or performing any suitable identification and/or recognition.

Third, variants of the method can improve customer relations management (CRM) for insurance entities (e.g., companies, agencies, underwriters, etc.), for whom FNOL generation and receipt can be a time- and resource-intensive, customer-facing process. For example, the method can include providing insurance claim data (e.g., an FNOL) to an insurance entity (e.g., a claims agent, an insurance provider, a representative of an insurance provider, etc.) automatically, and with relevant vehicle event data (e.g., collision data, fault statuses of one or more participants in a collision or other loss event, etc.), with or without a human manually submitting a claim (e.g., approving an FNOL submission, providing claim data to an application, etc.). In such variants, the method can enable insurance entities to reduce claims processing time by having the requisite data for claim resolution provided automatically, and/or to reduce follow-up time to ensure accurate insurance claim data. In another example, the method can include comparing manually submitted claims data (e.g., by an collision or adverse event participant)

Fourth, variants of the method can include automatically reporting vehicle events to various services (e.g., vehicle-history reporting services, emergency services, insurance claim adjusters, etc.). The method can provide rich data to such services that enable the services to modify their responses to a vehicle event (e.g., crash severity, estimation of injury type and likelihood, vehicle telemetry data, etc.).

Fifth, variants of the method can enable interaction between an onboard vehicle system and a user device (e.g., a mobile device of the user, a portable computer of the user, etc.), which can facilitate efficient collection and reporting of insurance claim data. For example, the onboard vehicle system can sample sensor data (e.g., of onboard vehicle

4

system sensors) and process the sampled sensor data to determine an adverse vehicle event has occurred, automatically populate a report with insurance claim data and tag the report with sensor data (e.g., video data) from the adverse vehicle event, and transmit the report to a user device. At the user device, an application (e.g., a client application executing on the user device) can notify the user that an adverse event has occurred, request approval of the insurance claim data depicted by the report, and/or perform any other suitable action. In another example, the onboard vehicle system can utilize short-range communication protocols (e.g., Bluetooth, BLE, etc.) to transfer data to the user device, and the user device can utilize long-range communication protocols (e.g., cellular radio transmission, wireless internet transmission, etc.) to transfer data to a remote computing system; in such examples, the user device can function as a communication relay for the onboard vehicle system.

Sixth, variants of the method and system can enable determination of an adverse vehicle event using imagery exclusively (e.g., without auxiliary sensor input such as accelerometer data, audio data, etc.). For example, kinematic motion of the vehicle in which the onboard vehicle system is installed (e.g., the ego-vehicle) can be extracted from image data (e.g., video sequences, image streams, video streams, single image frames, etc.). Kinematic motion can include ego-vehicle acceleration and deceleration, attitudinal shift (e.g., heading change, orientation change, etc.), impact parameters (e.g., jerk, whiplash, bounce, impulse, and other parameters during or in relation to impact of an object with the ego-vehicle, etc.), and any other parameters or values that describe the motion of the vehicle. Using imagery can enable an image processing module to be used to determine vehicle event data, and the image processing module can be locally or remotely located (e.g., and the same or substantially similar processed output can be obtained). In examples, a population of vehicles having corresponding onboard vehicle systems, wherein each onboard vehicle system includes a substantially identical camera model (e.g., a camera defining the same or substantially similar operating parameters such as resolution, effective shutter speed, color response, etc.), can generate imagery data gathered under various vehicle operating conditions that can then be analyzed using a centralized image processing module without the need to account for differing vehicle dynamics (e.g., vehicle dynamics may not affect relevant properties of the image data, whereas vehicle dynamics may affect signal characteristics of inertial measurements). In related examples, image data can be similarly unaffected by vehicle-specific dynamics in determining vehicle event data (e.g., adverse vehicle event data) for any individual vehicle (e.g., ego-vehicle).

Seventh, variants of the method allow insurance entities (e.g., insurance providers) to automatically or manually pull information supporting a claim (e.g., video collected before, after, and/or during a collision) from the system (e.g., from remote computing system 800). In one example, the insurance entities can request supporting information from the system, wherein the request can include: a collision identifier (e.g., vehicle identifier, timestamp, geolocation, driver identifier), an access credential (e.g., an API key assigned to the entity), the type of requested information, and/or any other suitable information.

However, the method, related systems, and variants thereof can additionally or alternatively afford any other suitable benefits and/or advantages.

3. Method

The method includes Block S100, which includes determining vehicle event data. Block S100 functions to obtain

5

data (e.g., information, signals, sensor data, etc.) related to a vehicle event (e.g., describing a vehicle event, during a vehicle event, subsequent to a vehicle event, prior to an imminent vehicle event, etc.) for use in other Blocks of the method. Block S100 can also function to monitor vehicle operation and/or driver behavior. Block S100 can also function to track vehicle usage. Determining vehicle data can include collecting vehicle sensor data, collecting vehicle identification data, determining driver behavior data, and determining occupant data, as well as any other suitable sub-blocks. Block S100 can include extracting vehicle event data from sensor data (e.g., via analyzing sensor data to determine vehicle event parameters, event parameters, etc.). Sub-blocks of Block S100 can be concurrently, sequentially, asynchronously, and/or otherwise suitably performed with any suitable temporal characteristics.

Block S100 is preferably performed by an onboard vehicle system coupled to the vehicle. The onboard vehicle system can include one or more cameras, an accelerometer, an inertial measurement unit (IMU), a gyroscope, a GPS module, and any other suitable sensors. However, the onboard vehicle system can additionally or alternatively include any suitable sensors, and omit any suitable sensors. In an example, the onboard vehicle system includes an imaging module that includes an interior-facing camera and an exterior-facing camera, wherein the imaging module is installed at the dashboard of the vehicle (e.g., coupled to the windshield proximal the dashboard, attached to a support member of the dashboard, mounted to the windshield proximal the rearview mirror, etc.). In this example, the onboard vehicle system also includes a plurality of IMUs and a wireless communication link (e.g., to a mobile device associated with a vehicle occupant, to an onboard diagnostic module/OBD module coupled to the vehicle, etc.). In another example, the onboard vehicle system includes only image sensors (e.g., cameras, optical sensors, etc., without any inertial or other auxiliary sensors), a wireless communication link, and a processing system (e.g., including a processor, onboard memory, any other suitable computing components, etc.). However, the onboard vehicle system can include any other suitable components.

Determining vehicle event data is preferably performed substantially continuously (e.g., continuously sampling). For example, the onboard vehicle system of the vehicle can continuously record interior and exterior video imagery and maintain a rolling buffer (e.g., a 10 minute buffer, a 20 minute buffer, a five hour buffer, an adjustable buffer, any suitable buffer, etc.). In a related example, determining vehicle event data can include storing gathered sensor data (e.g., losslessly storing uncompressed data, storing in a lossy manner via compression, storing all gathered sensor data, storing some gathered sensor data, etc.) on-board the vehicle, and/or transmitting sensor data (e.g., video imagery) off-board the vehicle. In alternatives, determining vehicle event data can be performed in response to user input (e.g., in response to activating a record button of the onboard vehicle system), periodically (e.g., once per minute, once per second, any suitable periodicity), and/or asynchronously. However, determining vehicle event data can include gathering, transferring, and/or otherwise manipulating vehicle event data with any suitable temporal characteristics.

In relation to Block S100, sensor data can be captured and/or sampled in response to a trigger. For example, Block S100 can include sampling an accelerometer signal from an accelerometer of the onboard vehicle system (e.g., substantially continuously, periodically during vehicle operation, etc.), and triggering data capture (e.g., image data sampling,

6

video capture, etc.) when the accelerometer signal exceeds a threshold value (e.g., corresponding to a rapid acceleration, heading deviation, deceleration, etc.). The threshold value can be predetermined (e.g., based on historical behavior), dynamically determined (e.g., based on present vehicle speed and/or heading, based on present vehicle acceleration, etc.), and/or otherwise suitably determined.

Sampling data in response to a trigger can function to allow adverse vehicle events to be determined in unexpected situations (e.g., in environments that can be difficult to continuously monitor, when the ego-vehicle is behaving nominally and a secondary vehicle collides with it, etc.). For example, in variations wherein the onboard vehicle system includes an accelerometer, Block S100 can include sampling an accelerometer signal from the accelerometer at a time point prior to the time interval in which an adverse vehicle event occurs (e.g., a sideswipe of the ego-vehicle by a passing vehicle, a perpendicular collision of a secondary vehicle into the ego-vehicle in an intersection, etc.), and sampling the interior image data and the exterior image data in response to the accelerometer signal exceeding a threshold value (e.g., when onset of the unexpected adverse vehicle event causes the accelerometer signal to exceed a threshold value, rate of change, etc.). In such variations, the interior and/or exterior image data can be continuously collected and overwritten in a buffer and sampled from an interval based on the trigger (e.g., an accelerometer signal-based trigger, any other suitable trigger, etc.), collected asynchronously in response to the trigger (e.g., wherein image recordation occurs subsequent to trigger determination or receipt), and/or collected in any other suitable manner.

In a specific implementation in accordance with Block S100, the ego-vehicle can be parked (e.g., positioned in a parking space and stationary) and be involved in a collision with a moving secondary vehicle at a time point. Contemporaneously with the time point, the accelerometer signal can exceed a threshold value as a result of the collision (e.g., wherein the accelerometer is operable in a low-power standby mode and substantially instantaneously awakened by a movement-sensitive switch in response to the collision), and trigger image recordation by the onboard vehicle system (e.g., to enable participant identifiers, such as those corresponding to the secondary vehicle, to be determined, and for insurance claim data such as an FNOL to be automatically generated and transmitted to an endpoint such as an insurer of the ego-vehicle).

Collecting vehicle sensor data functions to obtain data from sensors of the vehicle (e.g., in-built sensors, third-party sensors). Collected sensor data can include video data (e.g., from a forward and/or rearward facing vehicle camera, a dashboard-mounted camera with inward and/or outward facing cameras, a windshield-mounted onboard vehicle system, etc.), inertial data, location data, and other suitable data. Collected sensor data can optionally include data (e.g., exterior video, kinematic data, etc.) collected by a second or third vehicle, such as the other vehicle in the collision or adverse event, a vehicle proximal the collision site, or any other suitable vehicle. Collected sensor data can also include extracted parameters and/or characteristics of video and other sensor data, such as information regarding nearby objects (e.g., object classification, object trajectories, speeds, velocities, etc.), elapsed driving time, and other suitable extracted data. Collected sensor data can also include environmental data (e.g., weather data, temperature data, precipitation data, wind speed data, visibility data,

etc.). However, collecting vehicle sensor data can otherwise include collecting any suitable sensor data in any suitable manner.

Collecting vehicle identification data functions to determine the unique vehicle(s) (e.g., participants, parties, etc.) involved in a vehicle event (e.g., an adverse vehicle event, a collision, etc.). The unique vehicles can include the vehicle in which the onboard vehicle system is installed (e.g., the ego-vehicle), but can additionally or alternatively include surrounding vehicles. Unique vehicles involved in a vehicle event can include event participants and/or identifiers thereof (e.g., participant identifiers). Vehicle identification data can include: a vehicle identification number (VIN), vehicle licensing information (e.g., license plate number, taxicab number, livery number, rideshare identification number, etc.), a vehicle maintenance history report, insurer information (e.g., insurer name, policy holder, policy underwriter, etc.), visually apparent vehicle features (e.g., vehicle color, vehicle markings, vehicle shape, visible vehicle components, etc.), and any other suitable data intrinsic to a particular vehicle or vehicles. Vehicle identification data can be collected in any suitable manner, including by associating a user device with vehicle identification data in a database (e.g., a remote database), receiving vehicle identification entered manually by a user (e.g., at an application executing on a mobile device of a user associated with the vehicle), retrieving vehicle identification from a remote database (e.g., from an API based on VIN lookup), identifying a vehicle identifier pre-associated with the onboard vehicle system, and/or in any other suitable manner.

In a specific example, Block S100 can include sampling sensor data over a time interval using an onboard vehicle system rigidly coupled to an ego-vehicle (e.g., to an inner surface of the windshield of the ego-vehicle). In a related specific example, Block S100 includes sampling interior image data over a time interval using an interior-facing camera of an onboard vehicle system rigidly coupled to an ego-vehicle, and contemporaneously sampling exterior image data over the time interval using an exterior-facing camera of the onboard vehicle system. In these examples, the image data is sampled from a monocular camera; however, in alternative examples, the image data can additionally or alternatively be sampled from any suitable camera or image sensor (e.g., a stereocamera, an ambient light sensor, a multi-spectral camera, etc.).

Collecting driver behavior data functions to obtain information indicative of driver behavior (e.g., driver of the vehicle) associated with a vehicular event. Driver behavior data can include video data (e.g., from a camera pointing inwards toward the occupants of the vehicle including the driver, passengers, etc., from a front facing camera, from an IMU, etc.), extracted parameters and/or characteristics of imagery and other data (e.g., distraction level from eye movements, distraction level from mobile device use frequency while vehicle is moving, tailgating behavior, etc.), and any other suitable data. In a specific example, determining driver behavior includes determining a behavioral risk score based on an analytical model of driver behavior correlated with adverse vehicle event occurrence (e.g., collected from instances of the method implemented on a population of vehicles and drivers, stored and retrieved historic data, etc.). However, collecting driver behavior data can include collecting any suitable data related to driver behaviors in any other suitable manner.

Collecting driver behavior data in accordance with variations of Block S100 can include determining driver behavior data based on sensor measurements (e.g., sampled sensor

streams, captured sensor data, etc.). In relation to collecting driver behavior data, Block S100 can include extracting driver behavior data from image data. The driver behavior data can correspond to a driver of the ego-vehicle, a driver of another vehicle (e.g., a participant in an adverse vehicle event, a collision, etc.). In a first variation, driver behavior data is collected using one or more cameras, including an interior-facing camera. The interior camera's field of view preferably includes all or a portion of the driver's head, and can optionally include all or a portion of the passenger-side volume (e.g., the passenger's body or head), all or a portion of the rear seats, all or a portion of the rear window, the driver's side window(s), the passenger side window(s), or include any other suitable portion of the vehicle interior. The image(s) can be used to optically identify and/or track the driver's head position, head movement, eye position (e.g., gaze direction), eye movement, parameters thereof (e.g., duration, angle, etc.), or determine any other suitable distraction factor value. The image(s) can additionally be used to determine the driving context, such as whether there is a passenger in the vehicle or where the passenger is located, the environment around the vehicle visible through the vehicle windows, and/or any other suitable context derivable from the image(s). The image(s) can additionally be used to verify or validate other determinations of driver behavior. For example, driver interaction with a user device (e.g., phone) can be verified when the user device is present in the concurrently recorded image. The image can be a visual spectrum image, hyperspectral image, IR image, or be sampled at any other suitable wavelength. The image can be recorded using ambient light, light emitted by a light emitter (e.g., from an LED flash, etc.), or using any other suitable light source. The image can be a stereoisimage (e.g., recorded by a stereocamera), a two dimensional image (e.g., recorded by a monocular camera), a single image, or be any other suitable image.

In a second variation of Block S100, the sensor measurements include cabin audio (e.g., vehicle interior noise), which can be sampled by the audio sensor arranged proximal the driver's volume (e.g., integrated with the onboard system, part of a user device associated with the driver, etc.) or otherwise sampled. The cabin audio can be used to infer the driver head position, the driving context (e.g., whether the driver is calm, whether there is a child in the vehicle, etc.), or be otherwise used. The cabin audio preferably includes driver audio, and can optionally include passenger-side audio, rear-seat audio, vehicle audio (e.g., generated by the vehicle), road noise, radio audio, notification audio, or any other suitable audio stream. The cabin audio can be a single stream (e.g., recorded by a single audio sensor), multiple streams (e.g., recorded by multiple collocated or distal audio sensors, wherein the cabin audio is recorded in stereo, etc.), or include any suitable number of audio streams.

In a third variation of Block S100, the sensor measurements include user device operation parameters, which can be determined by a user device associated with the user or otherwise determined. User device operation parameters can include the user device's: processing load (e.g., as determined from an activity monitor, the CPU load, the GPU load, etc.), processing load changes, power consumption, touch input operation, sensor measurements, or any other suitable set of operation parameters. User device sensor measurements can include user device kinematics, pressure, temperature, ambient light obstruction, or any other suitable parameter measurement indicative of driver interaction with the user device. The user device operation parameters can be

determined by one or more clients (e.g., applications, SDKs on multiple applications, etc.) running on the user device, or be otherwise determined. In one example of the variation, processing load increases can be indicative of user interaction with the user device. In a second example of the variation, touch input operation can be indicative of user entering data into the user device. In a third example of the variation, user device vibration can be indicative of the user device being knocked by the user. In a fourth example, user device translation can be indicative of the user device being lifted or picked up by the user. In a fifth example, an increase in the sampled user device ambient pressure or temperature can be indicative of the user device being held. However, any other suitable user device parameter can be monitored for user interaction with the user device.

In a fourth variation of Block S100, the sensor measurements include an image or video (e.g., set of images) of the vehicle exterior, which can be sampled by the outward-facing camera directed toward the front of the vehicle (e.g., through the front windshield) or otherwise sampled. The camera's field of view preferably includes all or a portion of the roadway oriented toward the front the vehicle (e.g., the foreground toward the front of the vehicle, the lane in which the vehicle is moving, the adjacent lane or lanes, etc.), and can optionally include the region(s) to the sides of the vehicle (e.g., the driver's side, the passenger's side), to the rear of the vehicle, above the vehicle (e.g., through a sunroof and/or moon-roof), or include any other suitable portion of the environment outside the vehicle. The image(s) can be used to optically identify and/or track the vehicle's position within the environment (e.g., localize the vehicle), position within a lane, position relative to other objects (e.g., vehicles, pedestrians, etc.), or determine any other suitable parameter value. The image(s) can additionally be used to determine the driving context, such as whether there is a second vehicle in front of or beside the vehicle in which the sensor is emplaced. The image(s) can additionally be used to determine, verify, and/or validate other determinations of driver behavior. For example, driver attentiveness can be measured based on an amount and/or frequency of lateral drifting within a lane of the roadway. Drifting within a lane can be based on vehicle path extraction (e.g., extraction of the vehicle trajectory from the image or images) and comparison to lane markers extracted from images of the road. The image can be a visual spectrum image, hyperspectral image, IR image, or be sampled at any other suitable wavelength. The image can be recorded using ambient light, light emitted by a light emitter (e.g., from an LED flash, etc.), or using any other suitable light source. The image can be a stereoisimage (e.g., recorded by a stereocamera), a single image, or be any other suitable image.

In a fifth variation of Block S100, the sensor measurements include a kinematic signal. The kinematic signal is preferably indicative of a kinematic variable of the vehicle (e.g., acceleration, velocity, position, jerk, etc.), but can additionally or alternatively be indicative of any suitable motion variable associated with the vehicle and/or vehicle occupants (e.g., the driver, a passenger, etc.). The kinematic signal is preferably collected by a kinematic measurement mechanism (e.g., inertial measurement unit) of the onboard system, but can additionally or alternatively be collected by any other suitable component. In a first example, the sensor measurements include an accelerometer output of an inertial measurement unit of the onboard system. However, the sensor measurements can otherwise include any suitable kinematic signal, vehicle information (e.g., steering wheel position, pedal position), indicator, or other information.

In variations, collecting driver behavior data and/or distraction data in accordance with Block S100 can include determining driver distraction substantially as described in U.S. patent application Ser. No. 15/805,348, filed 16 Jan. 2018, which is incorporated herein in its entirety by this reference. However, in additional or alternative variations, driver behavior and/or distraction can be otherwise suitably determined.

Determining occupant data functions to collect data that is intrinsic to the driver and/or passenger(s) of the vehicle. Determining occupant data can include determining occupant data based on image processing applied to imagery from an internal facing camera (e.g., performing facial recognition on an image of the cabin of the vehicle). In a specific example, image processing applied to imagery can include automatically anonymizing the image data using an iterative algorithm in which the image data is iteratively blurred for an increasing number of frames of a video sequence until facial recognition on the blurred image fails. Determining occupant data can also include counting a number of vehicle occupants (e.g., determining a number of occupants), counting a number of passengers of the vehicle (e.g., as distinct from a driver of the vehicle), determining the vehicle's or driver's insured status, or determining any other suitable data. Determining occupant data can include determining identifying data pertaining to occupants of the vehicle (e.g., of the driver), which can include a license number, a registration number, an occupant identity (e.g., name), other occupant identifying info (e.g., address, phone number, sex, gender, height, weight, eye color, hair color, etc.), insurance policy number, and/or any other suitable occupant identifying data. Occupant identifying data can be collected in any suitable manner, including by retrieving occupant identifying data from a database associated with a user account of the occupant, manually input by an occupant, automatically determined using facial recognition algorithms and associated third party databases (e.g., social media databases), and/or in any other suitable manner. Determining occupant data is preferably based on image data (e.g., sampled from one or more cameras of the onboard vehicle system), but can additionally or alternatively be based on other sensor data (e.g., such as via automatically determining a number of occupants based on a number of different voices detected in a sampled audio stream of cabin sound).

In relation to determining occupant data, Block S100 can include determining a number of occupants based extracted participant identifiers. In examples, extracting one or more participant identifiers can include extracting participant identifiers related to multiple participants within a single vehicle (e.g., occupants), and counting the number of unique participant identifiers determined (e.g., based on facial recognition of multiple faces in an interior image data frame). Additionally or alternatively, identifying characteristics besides the number of occupants can be determined (e.g., identifying data as described above). Determining occupant data is preferably performed within a time interval (e.g., contemporaneously with a time interval in which vehicle event data is determined), but can additionally or alternatively be performed outside the time interval (e.g., before a vehicle event occurs, proximal in time to the beginning of a driving session, or after a vehicle event occurs, immediately subsequent to the determination of an adverse vehicle event to determine who was involved in the adverse vehicle event occurrence, etc.), and/or at any other suitable time. Determining occupant data is preferably performed with respect to the ego-vehicle, but can additionally or alternatively be

11

performed in relation to other vehicles (e.g., a vehicle involved in a collision or other adverse vehicle event with the ego-vehicle); for example, Block S100 can include determining a number of occupants of a secondary vehicle that is involved in a head-on collision with the ego-vehicle, and utilize the number of occupants in further blocks of the method (e.g., in automatically contacting emergency services with a report of a total number of likely injured persons, demographic data regarding the injured persons, etc.).

In variations, Block S100 can include extracting vehicle event data from sensor data (e.g., image data). For example, Block S100 can include extracting vehicle event data from the interior image data and the exterior image data (e.g., using a processor of the onboard vehicle system). The vehicle event data thus extracted is preferably associated with a vehicle event occurring within a time interval (e.g., wherein the time interval is determined automatically in response to a trigger, wherein the time interval is a periodic time interval of any suitable duration, etc.); however, the event data can additionally or alternatively be associated with a vehicle event occurring outside the time interval and/or be extracted in response to a trigger (e.g., directly subsequent to trigger detection). Extracting the vehicle event data can include extracting a set of event parameters, wherein various event parameters define aspects (e.g., quantifiable aspects, qualitative aspects, etc.) of the vehicle event. For example, extracting the set of event parameters can include extracting: a plurality of participant identifiers (e.g., wherein each event participant identifier can be associated with insurance policy data corresponding to a participant in the vehicle event); an acceleration (e.g., average acceleration, peak acceleration, etc.) of the ego-vehicle (e.g., during the time interval); an attitudinal shift (e.g., peak attitudinal shift, average attitudinal shift, total orientation change, average angular motion, etc.) of the ego-vehicle (e.g., during the time interval); and any other parameters associated with the vehicle event.

Block S100 can include extracting event parameters from image data (e.g., exclusively from image data without any auxiliary data). For example, as shown in FIG. 5, Block S100 can include extracting the attitudinal shift (e.g., peak attitudinal shift) of the ego-vehicle by determining a first position of each object depicted in a first frame (e.g., of exterior image data, at a first time point, within a time interval, etc.), determining a second position of each object depicted in a second frame (e.g., of the exterior image data, at a second time point, subsequently within the time interval, etc.), and calculating the attitudinal shift of the ego-vehicle based on a comparison between the first position of each object and the second position of each object (e.g., by determining the magnitude of the shift of each object within the image frame, and calculating the peak attitudinal shift by taking the shift corresponding to the greatest magnitude out of each of the object). In this example, the position of each object can be defined in three dimensions (e.g., in three dimensional space surrounding the ego-vehicle) and extracted using a method for 3D reconstruction (e.g., a trained position classifier, an inverse rendering technique, etc.); however, the position of each object can additionally or alternatively be defined in two dimensions (e.g., relative to a boundary of the frame, relative to a projected plane on which the ego-vehicle is located, etc.) and the shift in position can be a projected shift (e.g., a projection of the three dimensional object displacement onto two dimensions) and/or otherwise suitably defined.

12

In another example of extracting event parameters from image data, Block S100 can include extracting the peak acceleration of the ego-vehicle by determining a first velocity of each object depicted in the first frame (e.g., of exterior image data, at a first time point, over a first sub-interval of a time interval defined by two or more frames, etc.); determining a second velocity of each object depicted in the second frame (e.g., of exterior image data, at a second time point, over a second sub-interval of a time interval subsequent to the first sub-interval, etc.); and, calculating the peak acceleration of the ego-vehicle based on a comparison between the first velocity of each object and the second velocity of each object. In this example, the velocity of each object wherein can be defined in three dimensions (e.g., relative to the ego-vehicle), and determined using a method for 3D time-resolved event reconstruction (e.g., as described above, using any other suitable technique, etc.); however, the velocity of each object can additionally or alternatively be defined in two dimensions (e.g., relative to a frame boundary, relative to a projected plane occupied by the ego-vehicle, etc.) and/or otherwise suitably defined.

The method includes Block S200, which recites: analyzing vehicle event data to generate an event analysis. Block S200 functions to determine whether a vehicle event (e.g., vehicular event) that occurred was an adverse vehicle event (e.g., a vehicle event resulting in damage, theft, and/or loss of the vehicle, injury to an occupant of the vehicle, a substantial likelihood of damage, theft, and/or loss of the vehicle, and/or a substantial likelihood of injury to an occupant of the vehicle, etc.), and can optionally classify the event, determine the severity of an adverse vehicle event, or determine other event characteristics. Block S200 can also function to reconstruct and/or extract aspects (e.g., characteristics, parameters) of the adverse vehicle event for use in subsequent Blocks of the method (e.g., vehicle orientations, vehicle trajectories, vehicle locations, the aforementioned information as a function of time and/or space, etc.). The event analysis generated as a result of analyzing the vehicle event data preferably includes the determination of whether the vehicle event was an adverse vehicle event, but can additionally or alternatively include any suitable determined attributes of the vehicle event data.

Block S200 is preferably performed by a remote computing system, and can thus include transmitting vehicle event data to a remote computing system for analysis. Alternatively, Block S200 can be performed in a distributed fashion, distributed between a remote computing system and a local computing system (e.g., a mobile device, a vehicle computer, vehicle computing unit, electronic control unit, etc.). In a specific example of Block S200, a local computing system at the vehicle selects vehicle event data based on matching a pattern exhibited by the vehicle event data with a corresponding pattern of a set of predetermined patterns, and transmits the selected data to a remote computing system that implements a set of rules (e.g., derived from a machine learning model) on the received selected data. However, Block S200 can be otherwise suitably performed in a contiguous and/or distributed manner at any suitable computing system(s) in any suitable location(s).

Block S200 is preferably performed continuously (e.g., in real- or near-real time as data is collected), but can additionally or alternatively be performed upon detection of a trigger event (e.g., a rapid change in vehicle telemetry), asynchronously, periodically, or with any other suitable temporal characteristics. Block S200 is preferably implemented, at least in part, using a synthetic neural network (e.g., a convolutional neural network/CNN), but can addi-

13

tionally or alternatively be wholly or partially implemented using a Bayesian, deterministic, probabilistic, stochastic, and/or classification-based model, and/or any other suitable computational model. In a first variation, Block S200 includes matching a combination of sampled measurement values to a predetermined pattern, wherein the predetermined pattern is indicative of an adverse vehicle event. In a specific example of this variation, Block S200 includes performing a curve fitting procedure on a curve of acceleration versus time to identify a set of curve features (e.g., peaks, troughs, slopes, etc.) known to correspond to a head-on collision (e.g., a curve that includes a rapid direction change with a crumple-zone reversal, as shown in FIG. 2). In a second variation, Block S200 includes determining that a first sensor measurement (e.g., an accelerometer measurement) satisfies a condition (e.g., surpasses a threshold value), triggering an analysis of a second sensor measurement (e.g., video data collected proximal in time to the first sensor measurement) upon determining that the first sensor measurement satisfies the condition, and determining that an adverse vehicle event has occurred based on the second sensor measurement. In a third variation, Block S200 includes continuously classifying a combination of sampled measurement values (e.g., obtained via a shifting sampling time window of any suitable time range), and determining that an adverse vehicle event has occurred and characteristics of the adverse vehicle event based on the combination of sampled measurement values. In a fourth variation, Block 200 includes ingesting vehicle sensor data (e.g., airbag sensors, etc.) and/or monitoring for vehicle subsystem operation associated with adverse events (e.g., airbag deployment). In a fifth variation, Block 200 includes detecting the adverse vehicle event from the external-facing video, such as by detecting an object less than a threshold distance from the vehicle or camera, detecting a newly crumpled hood in the frame, or detecting other visual fiducials. However, Block S200 can include otherwise analyzing the vehicle event data in any other suitable manner.

Block S200 can optionally include determining a false positive, which functions to identify vehicle events that were initially determined to be an adverse vehicle event but have a substantial likelihood of being a nominal vehicle event. For example, Block S200 can include implementing a set of rules using the sensor data as an input and extracting a probability that the sensor data corresponded to a nominal vehicle event (e.g., closing a trunk lid) and not to an adverse vehicle event, and determining that the vehicle event is not an adverse vehicle event based on the probability exceeding a predetermined probability threshold.

Block S200 can optionally include determining characteristics of the adverse vehicle event based on the analysis of the collected sensor data. Determining characteristics of the adverse vehicle event can include segmenting of vehicle event types (e.g., between major collisions and minor collisions, high speed collisions and low speed collisions, etc.) into any suitable number of segments. Determining characteristics can also include determining a likelihood (e.g., probability) of injury to vehicle occupants and/or other entities involved in an adverse vehicle event. A likelihood of injury can be determined based on any suitable vehicle data, such as data indicative of airbag deployment (e.g., coupled with determined occupant bodyweight and/or height data), occupant medical history, a predictive model that associates vehicle event data with injury type and/or likelihood, and/or any other suitable data.

Block S200 can include determining a fault status associated with the driver. The fault status preferably defines the

14

responsibility of one or more participants in an adverse vehicle event; for example, Block S200 can include determining that a driver that rear ends a leading vehicle without any extenuating circumstances (e.g., such as the leading vehicle driving unsafely) is “solely at fault.” The fault status can include: sole responsibility, wherein a single party or participant bears total fault for the occurrence of an adverse event; partial responsibility (e.g., wherein the driver failed to take reasonable actions that may have reduced a severity of an adverse event but did not cause the adverse event, wherein two or more drivers independently took actions that together caused a collision or other adverse vehicle event, etc.); no-fault or no responsibility (e.g., wherein the driver could not reasonably have taken any action to avoid or reduce the severity of an adverse vehicle event); and any other suitable designation of fault or responsibility. The determination of the fault status is preferably based on driver behavior data (e.g., extracted or otherwise suitably determined in one or more variations of Block S100); however, data on which to base the determination of fault status can be any other suitable data (e.g., collected in accordance with one or more variations of Block S100).

A specific implementation of Block S200 includes detecting a trigger (e.g., a velocity change exceeding a predetermined threshold), uploading vehicle event data to a remote computing system, and analyzing the uploaded vehicle event data at the remote computing system. Detecting the trigger can include determining a time period roughly centered at the time point of trigger detection, and selecting the vehicle data determined in Block S100 within the determined time period. The time period can be an adjustable time period (e.g., based on queries submitted at a later time point from an insurance claim entity such as an agent, after reviewing received insurance claim data), a fixed time period, and/or any other suitable time period. In a first variation of this specific implementation, uploading the vehicle event data can be based on the detected trigger (e.g., uploading when the trigger is detected). In a second variation of this specific implementation, uploading the vehicle event data can be performed continuously (e.g., can include continuously streaming vehicle data to a remote computing system). However, uploading can be otherwise performed with any suitable temporal characteristics (e.g., asynchronously, periodically, etc.). In this specific implementation, analyzing the uploaded vehicle event data can include analyzing inertial data (e.g., gathered using an IMU of an installable onboard vehicle system) and video data (e.g., gathered using one or more cameras of an installable onboard vehicle system). Analyzing the uploaded vehicle event data can optionally include determining a false positive, and can further include storing the contextual data surrounding the false positive for subsequent retrieval and future false positive determination.

Block S200 can include determining a loss probability based on one or more event parameters (e.g., extracted or otherwise suitably determined in accordance with one or more variations of Block S100). In variations, Block S200 can include calculating a loss probability based on an ego-vehicle acceleration and an ego-vehicle attitudinal shift (e.g., extracted from sensor data, determined based on image data, determined in accordance with one or more variations of Block S100, etc.). In related variations, Block S200 can additionally or alternatively include calculating a loss probability based on acceleration and/or attitudinal shift of secondary vehicles that are participants in the adverse vehicle event (e.g., denoted by participant identifiers).

In variations, calculating the loss probability can be performed using an adverse vehicle event model that

15

receives an acceleration (e.g., a peak acceleration, an average acceleration, etc.) and/or an attitudinal shift (e.g., a peak attitudinal shift, an average angular orientation change over a time interval, etc.) as inputs and produces a probability of property loss as an output (e.g., a loss probability) based on insurance claim data (e.g., historical data, gathered population data, etc.). The model can additionally or alternatively produce a probability-weighted cost estimate as an output (e.g., an expected value of a repair and/or replacement cost). The adverse vehicle event model can be a trained machine learning model, wherein the model is trained based on a corpus of event parameters (e.g., extracted from vehicle event data corresponding to vehicle events experienced by a population of ego-vehicles, as well as corresponding insurance claim data associated with the events experienced by the population of ego-vehicles.). In one example, the adverse vehicle event model can be trained on claim outcomes pulled from an insurance provider (e.g., total cost, such as losses or expenses; time to closure; cost by category, such as personal injury or property damage; etc.), wherein the adverse vehicle event model can predict loss amounts based on collision type (e.g., class) and collision context (e.g., driving context). In a specific example, the adverse vehicle event model can identify historic claims with similar collision type and collision context, and determine a predicted loss for an instantaneous claim based on the similar historic claims. The loss type associated with the loss probability can be property loss (e.g., vehicle loss, vehicle damage, objects damaged by the vehicle over the course of the adverse event, etc.), personal injury loss (e.g., occupant death, occupant injury, etc.), and any other suitable loss. However, the adverse vehicle event model can additionally or alternatively include any other suitable model (e.g., a deterministic model, a set of equations, a probabilistic model, a lookup table, etc.) otherwise implemented in any other suitable manner.

The method includes Block S300, which recites: automatically generating insurance claim data based on the determined adverse vehicle event. Block S300 functions to organize information necessary for preparing and/or filing (e.g., submitting) an insurance claim to an insurance entity (e.g., an insurance underwriter, an insurance agency, an insurance agent, a third party claims administrator, any other suitable entities related to the main customer insurance entity, etc.). Block S300 can also function to provide rich data (e.g., audio, video, environmental, dynamic, static, and/or any other suitable data) associated with an adverse vehicle event to enable claim value (e.g., monetary value) to be accurately assessed by a user (e.g., an insurance claim agent). Block S300 can include compiling vehicle event data, compiling auxiliary data, and estimating claim costs (e.g., monetary costs). Block S300 is preferably performed by a remote computing system (e.g., the remote computing system by which Block S200 can be performed), but can additionally or alternatively be performed by the vehicle computing system (e.g., vehicle computer, processor of a mobile device associated with the vehicle and/or a vehicle occupant, etc.). Block S300 can include retrieving any suitable data (e.g., vehicle event data, intrinsic vehicle data, occupant data, etc.) from any suitable onboard vehicle system and/or computing system associated with the vehicle. In further alternatives, Block S300 can be performed in a distributed fashion (e.g., an insurance claim data template can be generated at a vehicle computing system and populated at a remote computing system, or vice versa), and/or otherwise performed at any suitable location(s).

16

In one variation, Block 300 includes: classifying the vehicle event (e.g., degree of severity, whether the event or event class requires a human claims adjuster or can be sent to an automated adjuster system, the fault status, the injury probability, etc.), automatically selecting one or more insurance claims (e.g., insurance forms) to fill out based on the vehicle event class, automatically selecting an endpoint for the insurance claim, and automatically transmitting the insurance claim to the endpoint. In a specific example, the method can include selecting, filling out, and sending a first insurance claim to an automatic claims system in response to vehicle event classification as a first class associated with automatic claims adjustment (e.g., a rear-end collision with known insurers), and include selecting, filling out, and sending a second insurance claim to a human claims system in response to vehicle event classification as a second class associated with human claims adjustment (e.g., a hit-and-run collision). However, the claims can be otherwise determined and routed.

Compiling vehicle event data functions to organize the determined characteristics of the adverse vehicle event determined in Block S200. Compiling the vehicle event data can include retrieving video data of the vehicle exterior, vehicle interior, and/or any other suitable location collected by the onboard vehicle system proximal in time to a time point of the determined adverse vehicle event. Compiling the vehicle event data can also include retrieving some or all the telemetry data collected proximal in time to the time point of the determined adverse vehicle event (e.g., OBD II port data), retrieving the insurance number (e.g., policy number, insured name, etc.). However, compiling vehicle event data can include receiving, retrieving, and/or otherwise suitably determining the vehicle event data in any suitable manner.

Compiling auxiliary data functions to organize other data related to the vehicle and/or other entity associated with the determined adverse vehicle event. Auxiliary data can include vehicle data (e.g., as described above in relation to Blocks S100 and S200), occupant data (e.g., as described above in relation to Blocks S100 and S200), and any other suitable data (e.g., location data, timestamp, historical data, etc.) associated with one or more characteristics of the adverse vehicle event (e.g., such as a medical history of an occupant, to be combined with an estimated crash severity to determine the likelihood of a legitimate medical injury claim; a maintenance history of the vehicle, to be combined with a determined component failure that caused an adverse vehicle event; visual data characterizing injury of the driver and/or other occupant, to be combined with determined severity of the vehicle event and/or a responsive action to contact emergency services; any other suitable characteristics; etc.).

Estimating claim costs functions to determine, based on the adverse vehicle event determined in Block S200, the expected cost related to the adverse vehicle event. The expected cost can include monetary costs, such as those associated with repair, medical treatment, and any other suitable costs associated with adverse vehicle events. The expected cost can include time cost, such as an expected repair time (e.g., a time period wherein the vehicle will be inoperable), treatment time (e.g., an expected healing time for an injury type determined to be likely in Block S200), and any other suitable time costs associated with adverse vehicle events. The expected costs can be estimated based on models (e.g., one or more machine learning models) that accept vehicle event data and auxiliary data as inputs, a lookup table, a deterministic set of rules, and/or on any other

suitable basis. In variations, the estimation models can be trained on historic insurance claim data, claim adjuster labelled vehicle event data, and/or any other suitable training data. In a specific example, Block S300 can include estimating a claim cost based on the loss probability and the fault status (e.g., determined in accordance with one or more variations of Block S200). In variations, Block S300 can include estimating a claim cost based at least one of the loss probability, the fault status, the presence of injury (e.g., of one or more occupants of the vehicle, the driver, etc.), probability of litigation, probability of claim challenge (e.g., in a court, by an insurer, by any other related entity, etc.), and any other suitable basis. However, Block S300 can additionally or alternatively include otherwise suitably estimating the monetary cost of an insurance claim, an FNOL, and/or associated insurance claim data.

Block S300 can be performed in response to a loss probability (e.g., determined in accordance with one or more variations of Block S200) exceeding a threshold probability. In variations, Block S300 can include transforming the set of event parameters and the fault status for each of the plurality of participant identifiers into insurance claim data in response to the loss probability exceeding a threshold probability. In a specific example, transforming the event parameters and the fault statuses into the insurance claim data can include: determining that the event parameters correspond to an adverse vehicle event that includes a totaling of the vehicle (e.g., based on an estimated claim cost, determined based on the loss probability) and a “no-fault” fault status of the vehicle driver, and generating an insurance claim that indicates a totaling of the insured vehicle and that the driver was not at fault.

In examples, Block S300 and/or other portions of the method (e.g., Block S100, Block S200, etc.) can employ machine learning approaches including any one or more of: supervised learning (e.g., using logistic regression, using back propagation neural networks, using random forests, decision trees, etc.), unsupervised learning (e.g., using an a priori algorithm, using K-means clustering), semi-supervised learning, reinforcement learning (e.g., using a Q-learning algorithm, using temporal difference learning), a regression algorithm (e.g., ordinary least squares, logistic regression, stepwise regression, multivariate adaptive regression splines, locally estimated scatterplot smoothing, etc.), an instance-based method (e.g., k-nearest neighbor, learning vector quantization, self-organizing map, etc.), a regularization method (e.g., ridge regression, least absolute shrinkage and selection operator, elastic net, etc.), a decision tree learning method (e.g., classification and regression tree, iterative dichotomiser 3, C4.5, chi-squared automatic interaction detection, decision stump, random forest, multivariate adaptive regression splines, gradient boosting machines, etc.), a Bayesian method (e.g., naïve Bayes, averaged one-dependence estimators, Bayesian belief network, etc.), a kernel method (e.g., a support vector machine, a radial basis function, a linear discriminate analysis, etc.), a clustering method (e.g., k-means clustering, expectation maximization, etc.), an associated rule learning algorithm (e.g., an a priori algorithm, an Eclat algorithm, etc.), an artificial neural network model (e.g., a Perceptron method, a back-propagation method, a Hopfield network method, a self-organizing map method, a learning vector quantization method, etc.), a deep learning algorithm (e.g., a restricted Boltzmann machine, a deep belief network method, a convolution network method, a stacked auto-encoder method, etc.), a dimensionality reduction method (e.g., principal component analysis, partial least squares regression, Sammon mapping,

multidimensional scaling, projection pursuit, etc.), an ensemble method (e.g., boosting, bootstrapped aggregation, AdaBoost, stacked generalization, gradient boosting machine method, random forest method, etc.), and/or any suitable form of machine learning algorithm.

In examples, the method 100 includes training a claims adjustment module based on vehicle event data extracted from interior image data and exterior image data (e.g., sampled by one or more onboard vehicle systems) corresponding to a population of ego-vehicles (e.g., a corpus of event parameters). In such examples, transforming the set of event parameters (e.g., and the fault status for each of the plurality of participant identifiers) into insurance claim data is preferably performed using the claim adjustment module (e.g., trained using any suitable method of those described above and/or any other suitable method); however, transforming the set of event parameters into insurance claim data can be otherwise suitably performed.

The method includes Block S400, which recites: transmitting insurance claim data to an endpoint. Block S400 functions to make claim data accessible to external entities (e.g., that require data to initiate and/or track an insurance claim). Block S400 can also function to automatically maintain a record (e.g., a remotely stored record) of vehicle events (e.g., nominal vehicle events, adverse vehicle events). The endpoint is preferably a web service endpoint (e.g., defined by a WSDL document), but can additionally or alternatively be any suitable remote repository of information that is accessible by insurance-related entities (e.g., a PDF or HTML form that can be submitted to an entity or other endpoint, a database of insurance claim data, etc.). Block S400 is preferably performed by a communication module associated with the vehicle; the communication module can be integrated with the vehicle (e.g., a wireless internet modem of the vehicle), part of a mobile device associated with the vehicle and/or an occupant of the vehicle (e.g., a cellular radio), part of a mobile device paired with a dashboard camera module emplaced in the vehicle, and/or any other suitable communication module. In alternatives, Block S400 can be performed in a distributed manner; for example, data can be transmitted to a central repository (e.g., an endpoint server, a remote server, a remote computing system, etc.) where it is linked together according to any suitable protocols and/or processes and made available for on-demand retrieval by an entity.

Block S400 can, in variations, include determining a destination entity (e.g., an insurance company, department within the insurance company, specific insurance adjuster, etc.) to which to transmit the insurance claim data (e.g., based on insurance company associated with the vehicle, associated with a user device, associated with the installable onboard vehicle system, etc.), wherein the endpoint is associated with the destination entity. The destination entity can be selected based on the event parameter or characteristic value (e.g., based on the event severity, event class, driver fault, etc.), based on the estimated claim value or probability of resolution in favor of one or more parties (e.g., claims with a high probability of being challenged sent to a first entity, claims with a high probability of settlement sent to a second entity, low payout claims sent to a third entity, high probability of resultant litigation sent to a fourth entity, one or more of the aforementioned associated probabilities sent to the same entity, etc.), or otherwise selected. A high probability can be determined based on a threshold probability (e.g., 51%, 80%, 95%, etc.), an expected value (e.g., obtained via integrating a probability distribution of various claim outcomes or resolutions), and/or otherwise suitably

determined. The destination entity can be selected using a lookup table (e.g., specified by the entity), a trained neural network (e.g., trained based on entity success rates for a given claim parameter, etc.), heuristics, deterministics, or otherwise determined. As shown in FIG. 4, Block S400 can also optionally include prompting the driver of the vehicle (e.g., at a mobile application rendered at a display of a mobile device of the driver) to approve submission of the insurance claim data, and transmitting the insurance claim data upon receipt of driver approval.

In a specific example, Block S400 includes prompting a user corresponding to one participant identifier (e.g., extracted in accordance with one or more variations of Block S100) to provide a transmission instruction at a mobile device associated with the user. In this example and in response to receiving the transmission instruction, Block S400 includes automatically transmitting the insurance claim data to an endpoint at a remote computing system. The endpoint can be of various types, such as a web service endpoint, a database endpoint, a user-defined endpoint, and any other suitable type.

The systems and methods of the preferred embodiment and variations thereof can be embodied and/or implemented at least in part as a machine configured to receive a computer-readable medium storing computer-readable instructions. The instructions are preferably executed by computer-executable components preferably integrated with the system and one or more portions of the processor and/or the controller 430. The computer-readable medium can be stored on any suitable computer-readable media such as RAMs, ROMs, flash memory, EEPROMs, optical devices (CD or DVD), hard drives, floppy drives, or any suitable device. The computer-executable component is preferably a general or application specific processor, but any suitable dedicated hardware or hardware/firmware combination device can alternatively or additionally execute the instructions.

Although omitted for conciseness, the preferred embodiments include every combination and permutation of the various system components and/or method blocks.

The FIGURES illustrate the architecture, functionality and operation of possible implementations of systems, methods and computer program products according to preferred embodiments, example configurations, and variations thereof. In this regard, each block in the flowchart or block diagrams may represent a module, segment, step, or portion of code, which comprises one or more executable instructions for implementing the specified logical function(s). It should also be noted that, in some alternative implementations, the functions noted in the block can occur out of the order noted in the FIGURES. For example, two blocks shown in succession may, in fact, be executed substantially concurrently, or the blocks may sometimes be executed in the reverse order, depending upon the functionality involved. It will also be noted that each block of the block diagrams and/or flowchart illustration, and combinations of blocks in the block diagrams and/or flowchart illustration, can be implemented by special purpose hardware-based systems that perform the specified functions or acts, or combinations of special purpose hardware and computer instructions.

As a person skilled in the art will recognize from the previous detailed description and from the figures and claims, modifications and changes can be made to the preferred embodiments of the invention without departing from the scope of this invention defined in the following claims.

The invention claimed is:

1. A method performed by an onboard vehicle system, comprising:

electronically obtaining interior image data over a time interval, the interior image data generated by an interior-facing camera of the onboard vehicle system coupled to an ego-vehicle;

electronically obtaining exterior image data over the time interval, the exterior image data generated by an exterior-facing camera of the onboard vehicle system coupled to the ego-vehicle, wherein the exterior image data comprises first image data of a first image frame, and second image data of a second image frame;

electronically determining, by a processor of the onboard vehicle system, a set of event parameters;

determining a loss probability by the processor of the onboard vehicle system, wherein the event parameters comprise an ego-vehicle acceleration and an ego-vehicle attitudinal shift, wherein the loss probability is determined based on the ego-vehicle acceleration and the ego-vehicle attitudinal shift, and wherein the ego-vehicle attitudinal shift is determined by the processor based on a first positional data associated with the first image frame captured at a first time point in the time interval, and a second positional data associated with the second image frame captured at a second time point subsequent to the first time point;

after the processor determines the loss probability, outputting information related to a collision involving the ego-vehicle during the time period for transmission by a wireless device to a receiving device, wherein the information comprises one or more of at least a part of the interior image data, at least a part of the exterior image data, the event parameters, or any combination of two or more of the foregoing.

2. The method of claim 1, wherein the processor comprises a module trained based on vehicle event data.

3. The method of claim 1, wherein the obtaining of the exterior image data is performed in response to a detection of a trigger.

4. The method of claim 1, further comprising determining driver behavior data based on at least a part of the interior image data.

5. The method of claim 1, further comprising determining a fault status based on at least a part of the interior image data.

6. The method of claim 1, further comprising determining a number of occupant(s) in the ego-vehicle.

7. The method of claim 1, further comprising receiving an instruction from a mobile device of a user, wherein the information is outputted in response to the instruction.

8. The method of claim 1, wherein the ego-vehicle acceleration comprises a pattern of acceleration over time, and wherein the method further comprises identifying one or more features from the pattern of the acceleration over time.

9. The method of claim 1, wherein the determining of the loss probability is performed using an event model, wherein the event model comprises a neural network model.

10. The method of claim 1, wherein the onboard vehicle system performing the method is different from a generic computer.

11. An onboard vehicle system for coupling with an ego-vehicle, comprising:

an interior-facing camera configured to generate and to electronically provide interior image data over a time interval while the onboard vehicle system is coupled with the ego-vehicle;

21

an exterior-facing camera configured to generate and to electronically provide exterior image data over the time interval while the onboard vehicle system is coupled with the ego-vehicle, the exterior image data comprising a first image data of a first image frame, and a second image data of a second image frame; and
 a processor configured to electronically determine a set of event parameters;
 wherein the processor is configured to determine a loss probability, wherein the event parameters comprise an ego-vehicle acceleration and an ego-vehicle attitudinal shift and wherein the processor is configured to determine the loss probability based on the ego-vehicle acceleration and the ego-vehicle attitudinal shift, and wherein the processor is configured to determine the ego-vehicle attitudinal shift based on a first positional data associated with the first image frame captured at a first time point in the time interval, and a second positional data associated with the second image frame captured at a second time point subsequent to the first time point; and
 wherein the processor is configured to, after determining the loss probability, output information related to a collision involving the ego-vehicle during the time period for transmission by a wireless device to a receiving device, wherein the information comprises one or more of at least a part of the interior image data, at least a part of the exterior image data, the event parameters, or any combination of two or more of the foregoing.

22

12. The onboard vehicle system of claim 11, wherein the processor comprises a module trained based on vehicle event data.

13. The onboard vehicle system of claim 11, wherein the processor is configured to obtain the exterior image data in response to a detection of a trigger.

14. The onboard vehicle system of claim 11, wherein the processor is configured to determine driver behavior data based on at least a part of the interior image data.

15. The onboard vehicle system of claim 14, wherein the processor is configured to determine a fault status based on the driver behavior data.

16. The onboard vehicle system of claim 11, wherein the processor is configured to determine a number of occupant (s) in the ego-vehicle.

17. The onboard vehicle system of claim 11, wherein the processor is configured to receive an instruction from a mobile device of a user, and wherein the processor is configured to output the information for transmission to the receiving device in response to the instruction.

18. The onboard vehicle system of claim 11, wherein the ego-vehicle acceleration comprises a pattern of acceleration over time, and wherein the processing unit processor is configured to identify one or more features from the pattern of the acceleration over time.

19. The onboard vehicle system of claim 11, wherein the processor is configured to determine the loss probability using an event model, wherein the event model comprising a neural network model.

20. The onboard vehicle system of claim 11, wherein the onboard vehicle system is different from a generic computer.

* * * * *